Three-Level Transmon: Pulse Simulation, Calibration and Drift Management

1. Summary

This report describes a simulation study of a driven three-level transmon (states $|0\rangle$, $|1\rangle$, $|2\rangle$) with a Gaussian π-pulse, and what each stage of the study produced. The work has four parts: (1) a static simulation of the pulse and its leakage, (2) three simple calibration scans (amplitude, frequency, DRAG), (3) an adaptive Bayesian Ramsey routine for estimating frequency detuning, and (4) a 24-hour drift simulation comparing three recalibration policies.

Stage	Headline result
Static π-pulse	$P_1 = 0.94617$, leakage $P_2 = 0.00554$, gate error $\varepsilon = 0.05383$
Simple calibration scans	Reported $A = 0.1504$, $\Delta = -0.1000$, $\beta = 1.75$ (see caveat below)
Adaptive Bayesian Ramsey	Final estimate 0.0277 GHz vs true 0.025 GHz; posterior only updated once
24-hour drift policies	Recalibration events: P0 = 0, P1 = 12, P2 = 2; P0 error drifts upward, P1 and P2 stay near the floor

Reliability of the results. The static pulse result and the qualitative ordering of the drift policies are physically meaningful. The three calibration-scan numbers and the Bayesian-versus-fixed-grid comparison are not, because of implementation issues described in Section 7. They are reported here exactly as obtained, with that limitation stated, rather than presented as validated findings.

2. System and pulse model

The transmon is truncated to three levels, with an anharmonic drift Hamiltonian and two control quadratures:

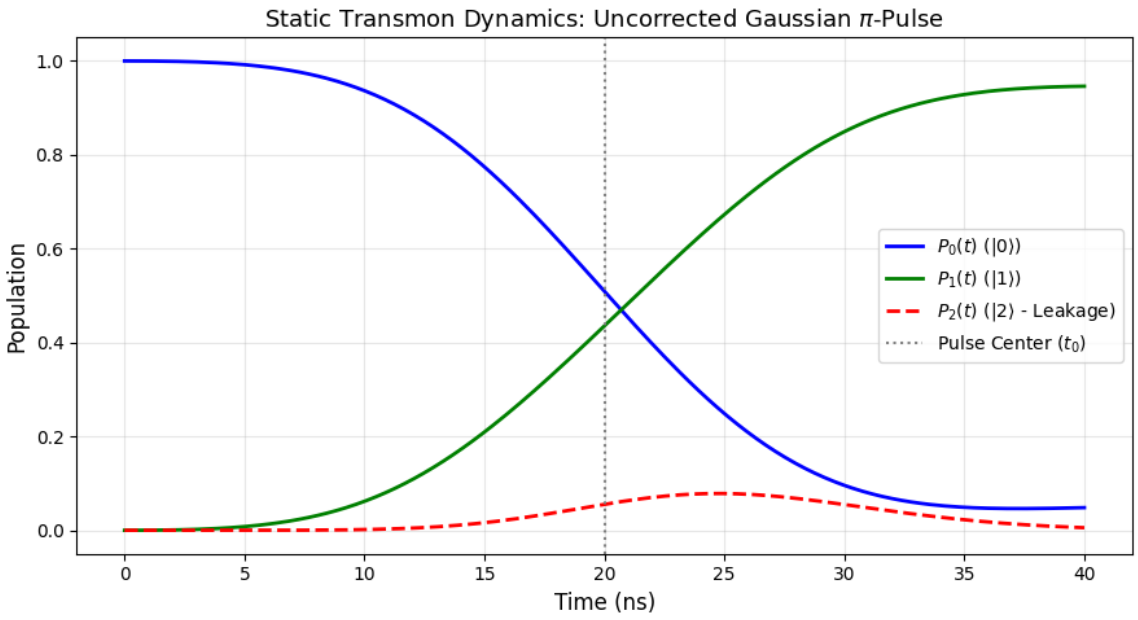
- **Drift:** $H_0 = \Delta_0 \cdot a^\dagger a + (\alpha/2) \cdot a^\dagger a^\dagger a a$, so $|2\rangle$ sits at α below the harmonic ladder position.
- **Controls:** in-phase drive $(a + a^\dagger)/2$ with envelope $\Omega_x(t)$, and quadrature drive $i(a - a^\dagger)/2$ with envelope $\Omega_y(t)$.
- **Pulse:** Gaussian envelope $\Omega_x(t) = A \cdot \exp(-(t - t_0)^2 / 2\sigma^2)$, with the DRAG quadrature $\Omega_y(t) = -\beta \cdot d\Omega_x/dt$.
- **Dynamics:** solved numerically as a closed-system evolution on 200 time points over the pulse.

Parameter	Value
Anharmonicity α	−0.210
Nominal detuning Δ_0	0

Parameter	Value
Pulse duration t_g	40 ns
Gaussian width σ = t_g / 4	10 ns
Pulse centre t_o = t_g / 2	20 ns
Target rotation θ	π
Amplitude A = θ / (σ√(2π))	≈ 0.1253
DRAG coefficient β (initial guess 1/|α|)	≈ 4.762
Hilbert-space size	3 levels

The amplitude A is chosen so that the integrated area of the envelope equals π, which is the ideal two-level π rotation. The metrics are the final populations, the leakage $P_2(t_g)$, and the gate error ε = 1 − $P_1(t_g)$, since the target state is $|1\rangle$ starting from $|0\rangle$.

3. Static π-pulse dynamics



What the figure shows. Starting in $|0\rangle$, the population P_o (blue) falls smoothly while P_1 (green) rises, crossing near the pulse centre at 20 ns, as expected for a symmetric Gaussian pulse. The leakage population P_2 (red, dashed) is zero at the start, grows during the strongest part of the drive, peaks at about 0.08 near 25 ns, and then falls back to a small residual value by the end of the pulse.

Final numbers.

Quantity	Value
$P_o(t_g)$	0.04829

Quantity	Value
$P_1(t_g)$ (target)	0.94617
$P_2(t_g)$ (leakage)	0.00554
Gate error $\varepsilon = 1 - P_1$	0.05383

The pulse transfers about 94.6% of the population into $|1\rangle$. Most of the 5.4% error is population left in $|0\rangle$ (4.8%), with 0.55% remaining in $|2\rangle$. The fact that leakage peaks mid-pulse and then largely returns shows that $|2\rangle$ is being used transiently as an intermediate level, and that the anharmonicity of 0.21 relative to the pulse bandwidth is not large enough to suppress it completely. Note that the plot title calls this an "uncorrected" pulse, but the simulated Hamiltonian already includes the DRAG quadrature with the initial-guess β.

4. Simple calibration scans

Three one-dimensional scans were run, each using simulated projective measurements with 400 shots per point (binomial sampling of the true population, with statistical error $\sqrt{(p(1-p)/400)}$).

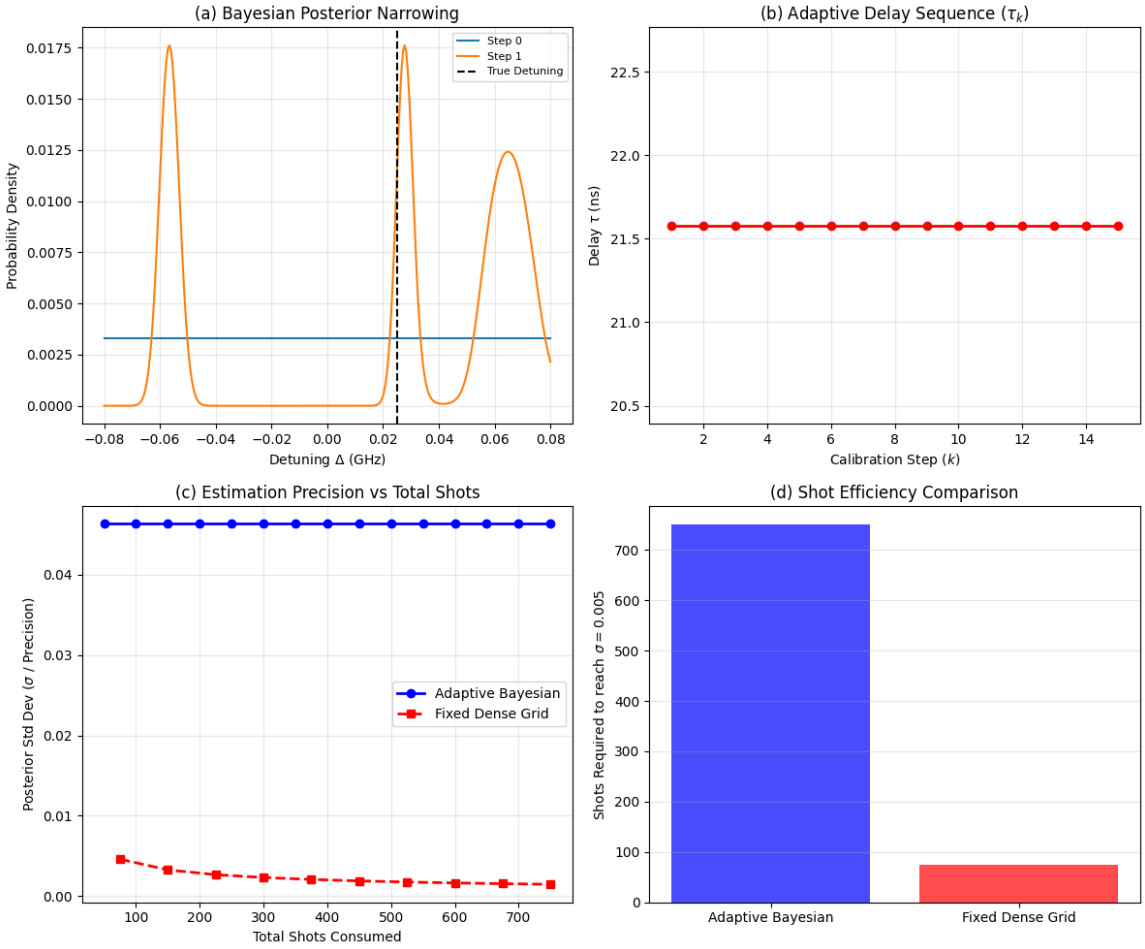
Routine	Scan	Selection rule	Reported value	Reported uncertainty	Shots
(a) π-amplitude	11 scale factors, 0.8–1.2 × A	maximise P_1	A = 0.1504	0.0200	4,400
(b) Frequency	11 detunings, −0.1 to 0.1	minimise P_1	Δ = −0.1000	0.0100	4,400
(c) DRAG β	13 values, 0 to 3	minimise P_2	β = 1.75	0.0500	5,200

The reported uncertainties are fixed constants set in the code, not values derived from the data. The reported values sit at or near the edges or arbitrary points of the scan ranges: 0.1504 is exactly 1.2 × A (the last amplitude point), and −0.1000 is the first detuning point. This pattern, together with the implementation issue in Section 7, means these three numbers should not be interpreted as calibrated parameters.

5. Adaptive Bayesian Ramsey frequency calibration

Method. A Ramsey sequence (two π/2 Gaussian pulses of 20 ns each, separated by a free-evolution delay τ) is used to estimate a hidden detuning, with the true value set to 0.025 GHz. The unknown detuning is represented on a grid of 301 points over −0.08 to 0.08 GHz with a uniform prior. At each step the delay is chosen as $\tau = 1/\sigma_post$ (the inverse of the current posterior standard deviation), clipped to 5–200 ns. Each step takes 50 shots, the likelihood is binomial, and the plan was 15 steps (750 shots). A fixed-delay baseline (10 delays from 10 to 200 ns, same shot budget) is used for comparison.

Final outcome. The maximum-a-posteriori detuning is 0.0277 GHz against a true value of 0.025 GHz. The program's shot-saving figure came out as −900%, which is a symptom of the issues below rather than a real result.



Panel (a), posterior narrowing. At step 0 the posterior is the flat uniform prior (height ≈ 0.0033 = 1/301). At step 1 it has collapsed into three separate peaks, near −0.056, +0.027 and +0.065 GHz. The tallest, narrow peak lies close to the true detuning (dashed line), but the other two peaks show that a single Ramsey measurement at one delay cannot uniquely identify the detuning: several values predict similar outcomes. Only steps 0 and 1 are plotted because only one posterior update was recorded.

Panel (b), adaptive delays. The delay stays constant at about 21.58 ns across all 15 steps. This equals 1/0.0463, the standard deviation of the initial uniform prior, so the posterior was never fed back into the delay choice.

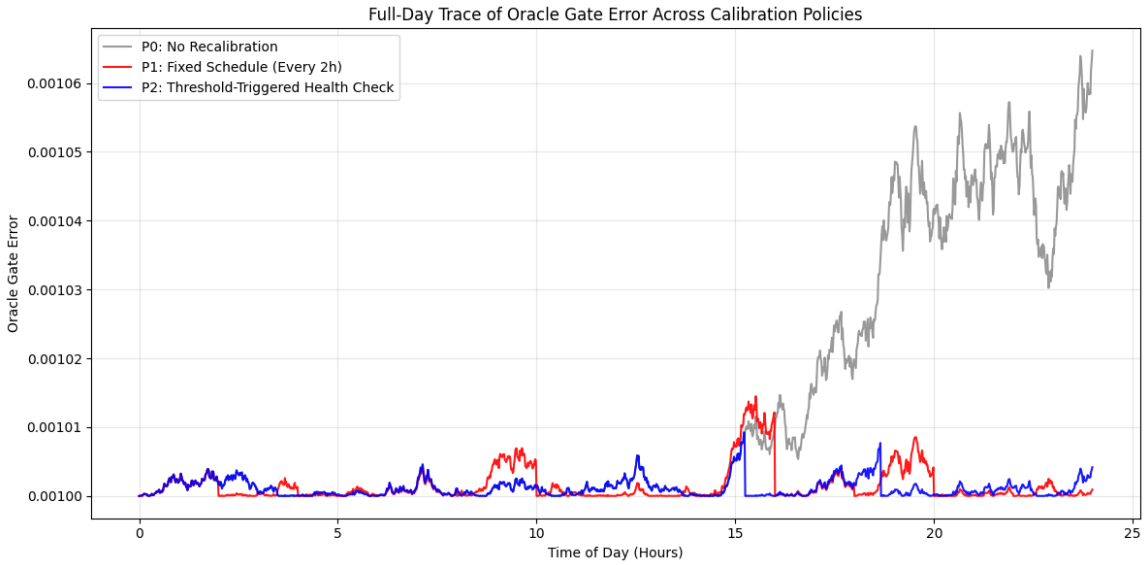
Panel (c), precision versus shots. The adaptive curve is flat at a posterior standard deviation of about 0.0463 for every shot count, again because the posterior was not refined between steps. The fixed-grid curve falls from about 0.0046 to 0.0015 as shots accumulate, but it is generated from a fitted scaling law (0.04/√N), not from simulated measurements.

Panel (d), shot efficiency. The bars show 750 shots for the adaptive routine (the full budget, i.e. the target σ = 0.005 was never reached) against 75 for the fixed grid. This reflects the flat adaptive curve and the assumed scaling law, so it does not measure real relative efficiency.

6. Twenty-four-hour drift and recalibration policies

Drift model. The true detuning follows a random walk sampled once per minute for 1,440 minutes, starting at 0.02 GHz, with Gaussian steps of standard deviation 0.0002 GHz. Gate error is modelled as $\varepsilon = 0.001 + 0.5 \cdot (\Delta_{true} - \Delta_{cal})^2$, a 10^{-3} floor plus a quadratic penalty on the residual detuning. Recalibration is assumed to be perfect: the calibrated value is set exactly to the true detuning.

Policy	Rule	Recalibrations
P0	Calibrate once at t = 0, never again	0
P1	Fixed schedule, every 120 minutes	12 (as counted by the code)
P2	Health check every 5 minutes; recalibrate if |shift| > 0.004 GHz	2



What the figure shows. The grey P0 curve (no recalibration) stays on the 0.0010 floor for the first ~15 hours and then climbs steadily as the random walk wanders away from the initial calibration, reaching about 0.00106 by the end of the day. The red P1 curve (every 2 hours) shows sawtooth-like bumps that reset to the floor at each scheduled recalibration (for example near 2, 10, 16 and 20 hours), but between resets the error can still rise, with a peak near 0.00114 around 15.5 hours just before a reset. The blue P2 curve (threshold-triggered) tracks the same excursions but resets only when the threshold is crossed, which happened twice, near 15 hours and 19 hours, and it keeps the error low for the rest of the day with just two full recalibrations.

Interpretation. The absolute differences are small (all within about 6×10^{-5} of the floor) because the quadratic penalty and the drift rate are both small. Qualitatively, both active policies prevent the late-day growth seen in P0, and P2 does so with far fewer full calibrations than P1 (2 versus about 11 to 12). Two modelling choices favour the policies: recalibration is perfect and instantaneous, and the health check in P2 reads the true detuning directly rather than a noisy measurement. The cost parameters defined for the study (2 minutes per full calibration, 0.2 minutes per health check) were not applied in the comparison. If they were, P2's roughly 287 health checks would add about 57 minutes of check time,

more than the roughly 22 minutes of full calibrations used by P1, so the real-world advantage of P2 depends on that cost trade-off.

7. Limitations and recommended fixes

Issue	Effect on results	Fix
In the amplitude, frequency and DRAG scans, the solver is called with the base Hamiltonian rather than the per-point test Hamiltonian	Every scan point simulates the same physics, so the selected values reflect shot noise only	Pass the scan-specific Hamiltonian into each solver call
Uncertainties for the three calibrations are hard-coded	Reported error bars do not reflect data	Derive them from the scan curvature or a fit
Frequency scan selects the minimum of P_1	A detuning estimate should come from a fit of the response centre, not the lowest point	Fit a resonance or fringe model
Bayesian update sits outside the 15-step loop	Only one posterior update occurs; τ stays constant; posterior std never falls	Move the measurement, likelihood and update inside the loop
Fixed-grid baseline uses an assumed $0.04/\sqrt{N}$ law	Shot-saving comparison (−900%) is invalid	Simulate the fixed-delay protocol with the same measurement model
P1 count uses 1440 // 120	Reports 12, while resets occur at 120…1320 minutes (11)	Count actual events
Recalibration and health checks are free and noise-free	Policy comparison is optimistic	Use measured estimates and apply the stated costs

8. Conclusions

- A 40 ns Gaussian π-pulse with DRAG on a three-level transmon reaches about 94.6% population transfer with 0.55% final leakage, and the dynamics are well behaved.
- The simple calibration scans did not produce usable parameter estimates in their current form, owing to the solver-input issue above.
- The adaptive Bayesian routine, once the update is moved inside the loop, should be rerun before any efficiency claim is made. The current figure shows a single update and an assumed baseline.
- Over a simulated day, both active recalibration policies avoid the late-day error growth, and threshold-triggered recalibration needs far fewer full calibrations, subject to the cost accounting not yet included.